

Safety Data Sheet according to UN-GHS

1. Identification

Product identifier:	MERACRYL® n-BMA
Chemical name:	n-Butyl methacrylate
Other means of identification	
Recommended use:	Monomer For use in industrial installations or professional treatment only.
Recommended restrictions:	Applications where liquid monomer is intended to come into contact with skin or nails.

Manufacturer/Importer/Distributor Information

Company Name	:	Roehm Hong Kong Co., Ltd. Unit 1504-05A, 15/F, 909 Cheung Sha Wan Road Cheung Sha Wan, Kowloon Hong Kong
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E-mail	:	SDS-PS-Asia@roehm.com

Emergency telephone number:

24-Hour Health Emergency	:	+86 532 8388 9090 (China mainland only) +49 6241 402 5280 (24h)
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2. Hazard(s) identification

Classification according to GHS

Physical Hazards

Flammable liquids	Category 3
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Health Hazards

Acute toxicity (Inhalativ)	Category 5
Skin irritation	Category 2
Skin sensitizer	Sub-category 1B
Specific Target Organ Toxicity - Single Exposure	Category 3 (Respiratory system)

Environmental Hazards

Acute hazards to the aquatic environment	Category 2
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Label Elements

Hazard Symbol:



Signal Word: Warning

Hazard Statement: Flammable liquid and vapor.
Causes skin irritation.
May cause an allergic skin reaction.
May be harmful if inhaled.
May cause respiratory irritation.
Toxic to aquatic life.

Precautionary Statements

Prevention: Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. Keep container tightly closed. Ground and bond container and receiving equipment. Use explosion-proof ventilating equipment. Use non-sparking tools. Take action to prevent static discharges. Avoid breathing dust/fume/gas/mist/vapors/spray. Wash thoroughly after handling. Use only outdoors or in a well-ventilated area. Contaminated work clothing should not be allowed out of the workplace. Avoid release to the environment. Wear protective gloves/eye protection/face protection.

Response: IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water [or shower]. Take off contaminated clothing and wash it before reuse. If skin irritation or rash occurs: Get medical advice/attention. IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a POISON CENTER or doctor/ physician if you feel unwell. In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish.

Storage: Store in a well-ventilated place. Keep cool. Keep container tightly closed. Store locked up.

Disposal: Dispose of contents/ container to an approved waste disposal plant.

Other hazards: Polymerization with heat evolution may occur in the presence of radical forming substances (e.g. peroxides), reducing substances, and/or heavy metal ions. Take precautionary measures against static discharges.

3. Composition/information on ingredients**Substances**

Chemical Identity	Common name and synonyms	CAS number	Content in percent (%) [*]
n-butyl methacrylate	butyl 2-methylprop-2-enoate	97-88-1	>99,5%

^{*} All concentrations are percent by weight unless ingredient is a gas. Gas concentrations are in percent by volume.

The exact concentration has been withheld as a trade secret.

4. First-aid measures

Description of first aid measures

General information:	First aider needs to protect himself. Take off all contaminated clothing immediately. Medical treatment is necessary if symptoms occur which are obviously caused by skin or eye contact with the product or by inhalation of its vapours.
Inhalation:	Move subject to fresh air and keep him calm. See a physician. If breathing is irregular or stopped, administer artificial respiration.
Skin Contact:	Wash off immediately with soap and water. If skin irritation occurs consult a physician.
Eye contact:	Flush eyes thoroughly with a large amount of water and consult a physician.
Ingestion:	Do not induce vomiting. Call a physician immediately.
Personal Protection for First-aid Responders:	First Aid responders should pay attention to self-protection and use the recommended protective clothing
Most important symptoms and effects, both acute and delayed Symptoms:	Headache.Dizziness.Causes skin and eye irritation.May cause an allergic skin reaction.
Hazards:	May cause sensitization by skin contact.May be harmful if inhaled.

Indication of immediate medical attention and special treatment needed

Treatment:	Treat symptomatically.
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5. Fire-fighting measures

General Fire Hazards:	Standard procedure for chemical fires. Keep out unprotected persons. Immediately evacuate personnel to safe areas. In case of fire, remove the endangered barrels and bring to a safe place, if this can be done safely. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. Water for fire fighting must not be introduced in the sewer system, subsoil, or surface waters. Assure that there are sufficient fire water retaining facilities Contaminated fire fighting water must be disposed of in conformity with the regulations of the local authorities.
Suitable (and unsuitable) extinguishing media	
Suitable extinguishing media:	Extinguish with alcohol-resistant foam, carbon dioxide or dry powder.
Unsuitable extinguishing media:	High volume water jet

Special hazards arising from the substance or mixture:

Closed container may rupture if strongly heated. Vapours may form explosive mixtures with air. May be released in case of fire: carbon monoxide, carbon dioxide, organic products of decomposition.

Special protective equipment and precautions for fire-fighters**Special fire-fighting procedures:**

Keep away from sources of ignition - No smoking. Take action to prevent static discharges. In the event of fire, cool the endangered containers with water. When heated above the flash point and/or during spraying (atomizing), ignitable mixtures may form in air. Use only explosion-proof equipment.

Special protective equipment for fire-fighters:

Wear self-contained breathing apparatus.

6. Accidental release measures**Personal precautions, protective equipment and emergency procedures:**

Assure sufficient ventilation. Use personal protective clothing. Keep away sources of ignition. Use breathing apparatus if exposed to vapours/dust/mist/aerosol. Do not breathe vapours or spray mist. Do not eat, drink or smoke when using this product. Wash hands thoroughly with soap and water after handling.

Accidental release measures:

Evacuate area and do not approach spilled product. ELIMINATE all ignition sources (no smoking, flares, sparks or flames in immediate area). For personal protection see section 8.

For emergency responders:

Avoid contact with eyes, skin, and clothing. Do not inhale vapours / aerosols. Observe regulations on prevention of water pollution (check, dam up, cover up).

Methods and material for containment and cleaning up:

Larger quantities: Remove mechanically (by pumping). Use explosion-proof equipment! Smaller quantities and/or residues: Contain with absorbent material (e.g. sand, diatomaceous earth, acid absorbent, universal absorbent or sawdust). Dispose of in accordance with regulations.

Environmental Precautions:

Prevent product from getting into drains/surface water/groundwater.

7. Handling and storage**Handling****Technical measures:**

Install appropriate equipment and wear appropriate personal protective equipment (see "8. Exposure control/personal protection").

Local/Total ventilation:

Use explosion-proof electrical, ventilating and lighting equipment.

Safe handling advice:

Keep container tightly closed. Ensure there is good room ventilation. Avoid breathing mists or vapors. Avoid contact with skin and eyes. Do not eat, drink or smoke during use. Do not get in eyes, on skin, on clothing. Keep away from

sources of ignition - No smoking. Take action to prevent static discharges. In the event of fire, cool the endangered containers with water. When heated above the flash point and/or during spraying (atomizing), ignitable mixtures may form in air. Use only explosion-proof equipment. Follow all SDS/label precautions even after container is emptied because it may retain product residues.

|| Contact avoidance measures:

see section 8. see section 10.

Storage**Safe storage conditions:**

Fill the container by approximately 90 % only as oxygen (air) is required for stabilisation. With large storage containers make sure the oxygen (air) supply is sufficient to ensure stability. Keep only in the original container at a temperature not exceeding 30 °C. Can polymerize with intense heat release. Protect from the action of light. See section 10: Materials to avoid Observe prohibition against storing together!

Safe packaging materials:

No data available.

8. Exposure controls/personal protection**Control Parameters****Occupational Exposure Limits**

Chemical Identity	Type	Exposure Limit Values	Source
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Please refer to the latest edition of the appropriate source text and consult an industrial hygienist or similar professional, or local agencies, for further information.

Observe national threshold limit values.

Biological Limit Values

No biological exposure limits noted for the ingredient(s).

Appropriate Engineering Controls

For monitoring procedures refer for instance to "Empfohlene Analysenverfahren für Arbeitsplatzmessungen", Schriftenreihe der Bundesanstalt für Arbeitsschutz and "NIOSH Manual of Analytical Methods", National Institute for Occupational Safety and Health

Individual protection measures, such as personal protective equipment**General information:**

No data available.

Eye/face protection:

tightly fitting goggles

Hand Protection:

Material: butyl rubber gloves (minimal thickness 0.3 mm)
Break-through time: 76 min
Guideline: EN 374
Additional Information: For each work-place a suitable glove type has to be selected., Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. Also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion, and the contact time., Gloves should be replaced regularly, especially after extended contact with the product., Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Other:

On handling of larger quantities: face mask, chemical-resistant boots and apron

Respiratory Protection:

Breathing apparatus in case of high concentrations

Hygiene measures:

Take off all contaminated clothing immediately. Store work clothing separately. Follow the usual good standards of occupational hygiene. Clean skin thoroughly after work; apply skin cream.

9. Physical and chemical properties**Information on basic physical and chemical properties****Appearance**

Physical state: liquid
Form: Liquid
Color: Colorless

Odor: ester-like

Odor Threshold: No data available.

Freezing point: -58 °F/-50 °C

Boiling Point: 325 °F/163 °C (1.013 hPa)

Flammability: Flammable liquid and vapor.

Upper/lower limit on flammability or explosive limits

Explosive limit - upper: 8 %(V)

Explosive limit - lower: 2 %(V)

Flash Point: 119,3 °F/48,5 °C (DIN 51755 / Abel Pensky Closed Cup)

Autoignition Temperature: 554 °F/290 °C Auto Ignition Temperature

Decomposition Temperature: Not applicable

pH: neutral in Water

Viscosity

Dynamic viscosity: No data available.

Kinematic viscosity: 0,81 mm²/s (104 °F/40 °C, DIN 51562)
1,06 mm²/s (68 °F/20 °C, DIN 51562)

Flow Time: No data available.

Solubility(ies)

Solubility in Water:	0,36 g/l (77 °F/25 °C, OECD 105) source: literature
Solubility (other):	miscible with most organic solvents
Partition coefficient (n-octanol/water):	3,03 (measured; OECD 107)
Vapor pressure:	2,12 hPa (68 °F/20 °C)
Relative density:	No data available.
Density:	0,896 g/cm ³ (68 °F/20 °C)
Bulk density:	No data available.
Relative vapor density:	4,9 68 °F/20 °C
Other information	
Explosive properties:	Not explosive Not to be expected in view of the structure Vapours may form explosive mixtures with air
Metal Corrosion:	Not corrosive to metals

10. Stability and reactivity

Reactivity:	Polymerisation can occur.
Chemical Stability:	No decomposition if used as directed. The product is normally supplied in a stabilized form. If the permissible storage period and/or storage temperature is exceeded, the product may polymerize with heat evolution.
Possibility of hazardous reactions:	Polymerization with heat evolution may occur in the presence of radical forming substances (e.g. peroxides), reducing substances, and/or heavy metal ions.
Conditions to avoid:	The product is normally supplied in a stabilized form. If the permissible storage period and/or storage temperature is exceeded, the product may polymerize with heat evolution.
Incompatible Materials:	Peroxides, amines, sulfur compounds, heavy metal ions, alkalis, reducing agents and oxidizing agents. Mineral Acid Free radical initiators. Reactions with strong oxidizing agents.
Hazardous Decomposition Products:	None when used as directed.

11. Toxicological information

General information:	The substance is rapidly metabolized
Information on toxicological effects	
Inhalation:	May be harmful if inhaled.
Skin Contact:	Prolonged or repeated contact may cause skin sensitization in susceptible individuals.
Eye contact:	May be slightly irritating to skin and eyes.

Ingestion: If handled correctly, not a relevant route of exposure. Information on effects are given below.

Symptoms related to the physical, chemical and toxicological characteristics

Inhalation: Respiratory tract irritation. Headache. Dizziness.

Skin Contact: Prolonged skin contact may cause redness and irritation.

Eye contact: Eye may become red, tear, and become painful.

Ingestion: If handled correctly, not a relevant route of exposure. Information on effects are given below.

Information on likely routes of exposure

Acute toxicity (list all possible routes of exposure)

Oral

Product: Not classified for acute toxicity based on available data.

Components:

n-butyl methacrylate LD 50 (Rat): > 2.000 mg/kg
Own test result., (limit test), No deaths observed.

Dermal

Product: Not classified for acute toxicity based on available data.

Components:

n-butyl methacrylate LD 50 (Rabbit): > 2.000 mg/kg
Own test result., (limit test), No deaths observed.

Inhalation

Product: Acute inhalation toxicity category 5 (UN-GHS)

Components:

n-butyl methacrylate LC 50 (Rat, 4 h): 29 mg/l low toxicity after single exposure;
Vapor/aerosol Not toxic after single exposure; Dust and mist, Not applicable

Repeated dose toxicity

Product: NOAEL (Rat, Inhalativ, 4 Weeks): 1,83 mg/l Findings: damage to the nasal mucosa

NOAEL (Rat, Oral, 90 d): 120 mg/kg systemic effects

Components:

n-butyl methacrylate NOAEL (Rat, Inhalativ, 4 Weeks): 1,83 mg/l Damage to the nasal mucosa.

NOAEL (Rat, Oral, 90 d): 120 mg/kg systemic effects

Skin Corrosion/Irritation

Product: Skin irritant Category 2 (UN-GHS)
If contact with skin is prolonged and/or frequent, irritations cannot be excluded.

Components:

n-butyl methacrylate (Rabbit): Irritating.

Serious Eye Damage/Eye Irritation

Product: , Not Classified

Components:

n-butyl methacrylate Not irritating OECD 405 , Rabbit:

Respiratory or Skin Sensitization

Product: Skin Sensitisation Category 1B (UN-GHS)

Components:

n-butyl methacrylate

Local Lymph Node Assay (LLNA), OECD 429 (Mouse): Skin sensitizer
Not classified for respiratory sensitization

Carcinogenicity**Product:**

Not classified

Components:

n-butyl methacrylate

Not classified no specific test data available no evidence for hazardous properties (structure-activity-relationships) (analogy)

Germ Cell Mutagenicity

Not classified

In vitro**Product:**

Based on available data, the classification criteria are not met.;

Components:

n-butyl methacrylate

gene mutation (OECD 471): negative
gene mutation (OECD 476): negative , Chinese hamster lung fibroblasts (V79)
Chromosomal aberration (OECD 473): negative , Chinese hamster lung cells

In vivo**Product:**

no evidence of mutagenic effects;

Components:

n-butyl methacrylate

Micronucleus test (OECD 474) Oral (Mouse): negative

Reproductive toxicity**Product:**

Not classified

Components:

n-butyl methacrylate

Not classified No indications of teratogenic effects in experimental animals.

Specific Target Organ Toxicity - Single Exposure**Product:**

Respiratory tract irritation.

Specific target organ toxicity – single exposure Category 3 (UN-GHS)

Components:

n-butyl methacrylate

Inhalation - vapor: Category 3 with respiratory tract irritation.

Specific Target Organ Toxicity - Repeated Exposure**Product:**

Based on available data, the classification criteria are not met.

Components:

n-butyl methacrylate

Not classified

Aspiration Hazard**Product:**

No aspiration toxicity classification

Components:

n-butyl methacrylate

Not classified

Information on health hazards**Other hazards****Product:**

Avoid contact with the skin and eyes and inhalation of the product vapours.;

12. Ecological information

Ecotoxicity:**Acute hazards to the aquatic environment:**

Fish

Product: LC 50 (Oryzias latipes (Orange-red killifish), 96 h): 5,57 mg/l

Components:

n-butyl methacrylate LC 50 (Oryzias latipes (Orange-red killifish), 96 h): 5,57 mg/l

Aquatic Invertebrates

Product: EC 50 (Daphnia magna (Water flea), 48 h): 25,4 mg/l

Components:

n-butyl methacrylate EC 50 (Daphnia magna (Water flea), 48 h): 25,4 mg/l

Toxicity to Aquatic Plants

Product: EC 50 (Selenastrum capricornutum (green algae), 72 h): 31,2 mg/l (OECD 201)

Components:

n-butyl methacrylate EC 50 (Selenastrum capricornutum (green algae), 72 h): 31,2 mg/l (OECD 201)

Toxicity to microorganisms

Product: EC10 (Pseudomonas putida, 18 h): 31,7 mg/l (Cell multiplication inhibition test)

Components:

n-butyl methacrylate EC10 (Pseudomonas putida, 18 h): 31,7 mg/l (Cell multiplication inhibition test)

Chronic hazards to the aquatic environment:

Fish

Product: No data available.

Components:

n-butyl methacrylate No data available.

Aquatic Invertebrates

Product: NOEC (Daphnia magna (Water flea), 21 d): 1,1 mg/l (OECD 202 part 2)

Components:

n-butyl methacrylate NOEC (Daphnia magna (Water flea), 21 d): 1,1 mg/l (OECD 202 part 2)

Toxicity to Aquatic Plants

Product: NOEC (Pseudokirchneriella subcapitata (green algae), 72 h): 24,8 mg/l (OECD 201)

Components:

n-butyl methacrylate NOEC (Pseudokirchneriella subcapitata (green algae), 72 h): 24,8 mg/l (OECD 201)

Toxicity to microorganisms

Product: EC10 (Pseudomonas putida, 18 h): 31,7 mg/l (Cell multiplication inhibition test)

Components:

n-butyl methacrylate EC10 (Pseudomonas putida, 18 h): 31,7 mg/l (Cell multiplication inhibition test)

Persistence and Degradability

Biodegradation

Product: 88 % (28 d, OECD 301 C), Readily biodegradable

Components:

n-butyl methacrylate 88 % (28 d, OECD 301 C), Readily biodegradable

BOD/COD Ratio

Product: No data available.

Components:

n-butyl methacrylate No data available.

Bioaccumulative potential**Bioconcentration Factor (BCF)**

Product: Substance is metabolised via standard pathways. Therefore, significant bioaccumulation is not expected.

Components:
n-butyl methacrylate Substance is metabolised via standard pathways. Therefore, significant bioaccumulation is not expected.

Partition Coefficient n-octanol / water (log Kow)

Product: Log Kow: 3,03 25 °C (measured; OECD 107)

Components:
n-butyl methacrylate Log Kow: 3,03 25 °C (OECD TG 107, measured)

Mobility in soil:

Product The substance evaporates gradually into the atmosphere from the surface of the water. If the substance does get into the environment, it tends to remain in the compartment it was discharged into. Substance may bind to solid soil phase, sediment or clarification sludge due to its adsorptive behaviour.

Components:
n-butyl methacrylate The substance evaporates gradually into the atmosphere from the surface of the water. If the substance does get into the environment, it tends to remain in the compartment it was discharged into. Substance may bind to solid soil phase, sediment or clarification sludge due to its adsorptive behaviour.

Results of PBT and vPvB assessment:

Product Not a PBT, vPvB substance as per the criteria of the REACH Ordinance.

Components:
n-butyl methacrylate Non-classified vPvB substance,
Non-classified PBT substance

Other adverse effects:**Other hazards**

Product: Prevent substance from entering soil, natural bodies of water and sewer systems. Photochemical degradation (air) takes place.

13. Disposal considerations

General information: Dispose of waste and residues in accordance with local authority requirements.

Disposal methods:

Waste is hazardous. It must be disposed of in accordance with the regulations after consultation of the competent local authorities and the disposal company in a suitable and licensed facility. Strictly controlled conditions during disposal or treatment to air, wastewater and waste. Do not add wastewater to a biological wastewater treatment plant. Bring wastewater containing AOX for professional disposal. The waste key number must be determined as per the European Waste Types List (decision on EU Waste Types List 2000/532/EC) in cooperation with the disposal firm / producing firm / official authority.

Contaminated Packaging:

Contaminated packaging should ideally be emptied; it can then be recycled after having been decontaminated. Packaging that cannot be cleaned should be disposed of professionally. Uncontaminated packaging may be taken for recycling.

14. Transport information**International Regulations****IATA-DGR**

UN/ID No.	: UN 2227
Proper shipping name	: n-Butyl methacrylate, stabilized
Class	: 3
Packing group	: III
Labels	: 3
Packing instruction (cargo aircraft)	: 366
Packing instruction (passenger aircraft)	: 355

IMDG-Code

UN number	: UN 2227
Proper shipping name	: n-BUTYL METHACRYLATE, STABILIZED
Class	: 3
Packing group	: III
Labels	: 3
EmS Code	: F-E, S-D
Marine pollutant	: no

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

15. Regulatory information**International regulations****Montreal protocol**

Not applicable

Stockholm convention

Not applicable

Rotterdam convention

Not applicable

Kyoto protocol

Not applicable

16. Other information, including date of preparation or last revision**Issue Date:** 20.03.2019**Version #:** 2.1**Source of information:**

relevant manuals and publications
own examinations
own toxicological and ecotoxicological studies
toxicological and ecotoxicological studies of other manufacturers
SIAR
OECD-SIDS
RTK public files

Further Information:

The product is normally supplied in a stabilized form. If the permissible storage period and/or storage temperature is exceeded, the product may polymerize with heat evolution.

Disclaimer:

This information and all further technical advice is based on our present knowledge and experience. However, it implies no liability or other legal responsibility on our part, including with regard to existing third party intellectual property rights, especially patent rights. In particular, no warranty, whether express or implied, or guarantee of product properties in the legal sense is intended or implied. We reserve the right to make any changes according to technological progress or further developments. The customer is not released from the obligation to conduct careful inspection and testing of incoming goods. Performance of the product described herein should be verified by testing, which should be carried out only by qualified experts in the sole responsibility of a customer. Reference to trade names used by other companies is neither a recommendation, nor does it imply that similar products could not be used.